

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and deployed **machine learning pipelines** using **Python (scikit-learn, pandas, NumPy)** and **AWS Lambda** to automate feature engineering and model inference across **PostgreSQL** data warehouses, reducing manual data processing time by **60%** and enabling real-time predictive scoring at scale.
- Built **supervised and unsupervised learning models** including classification, regression, and clustering algorithms, achieving **94% accuracy** on validation datasets through rigorous **hyperparameter tuning**, **cross-validation**, and **feature selection** using statistical methods and domain expertise.
- Designed and operationalized **end-to-end data pipelines** with **ETL orchestration** using **Python** and **SQL** to extract, transform, and load enterprise data from multiple sources into **data warehouses**, supporting **analytics models** and **business intelligence dashboards** for **250+** stakeholders.
- Implemented **model monitoring and evaluation frameworks** using **CloudWatch**, **Splunk**, and custom **Python dashboards** to track model performance metrics, data drift, and prediction accuracy in production, enabling proactive retraining cycles and **99.5% uptime** SLA compliance.
- Collaborated with cross-functional teams to translate business requirements into machine learning objectives, conducting **exploratory data analysis**, **A/B testing**, and **statistical hypothesis testing** to validate model assumptions and deliver data-driven insights that improved operational efficiency by **35%**.
- Leveraged **Power BI** and **Tableau** to create interactive dashboards and **data visualizations** that surface machine learning model outputs, prediction distributions, and business KPIs; automated reporting via **Power Query** and **DAX** measures, reducing reporting overhead by **45%**.

SKILLS

Machine Learning & AI – scikit-learn, supervised learning, unsupervised learning, classification, regression, clustering, hyperparameter tuning, cross-validation, feature selection, model evaluation, evaluation metrics.

Data Engineering & Analytics – Python (pandas, NumPy), SQL (PostgreSQL, MySQL, Oracle), data pipelines, ETL orchestration, data warehousing, exploratory data analysis, statistical analysis, hypothesis testing, A/B testing, time-series analysis.

Cloud & DevOps – AWS (Lambda, EC2, S3, CloudWatch), Docker, CI/CD (Jenkins), Git, Linux, shell scripting.

Business Intelligence & Visualization – Power BI, Tableau, data modelling, DAX, Power Query, interactive dashboards, data visualization, reporting automation.

Monitoring & Production Systems – CloudWatch, Splunk, SLA management, incident response, application monitoring, model monitoring, model drift detection.

Programming & Frameworks – Python, Java, Spring Boot, REST APIs, Flask, FastAPI, R, RStudio.

Professional Certifications & Credentials – MSc Data Analytics (NCI, Dublin), Bachelor of Engineering in Computer Science, IELTS Academic Band 7, Full Stack Java Spring Boot & Angular, Web Development (UC Davis), Python Programming (University of Michigan).

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Real-Time Fraud Detection & Anomaly Scoring Pipeline (HCL Technologies)

- Engineered an end-to-end **machine learning pipeline** leveraging **scikit-learn**, **pandas**, and **NumPy** to detect fraudulent transactions and anomalies in high-volume payment streams, achieving **96.2% precision** and **91.8% recall** on held-out test sets across **50M+ monthly transactions**.
- Implemented **supervised classification models** (Gradient Boosting, Random Forest, Logistic Regression) with automated **hyperparameter optimization** and **cross-validation** frameworks; deployed models via **AWS Lambda** to score transactions in **sub-100ms latency**, enabling real-time risk decisions.
- Constructed robust **ETL pipelines** using **Python** and **SQL** to ingest transaction data from **PostgreSQL** and **Oracle** databases, perform **feature engineering** across 50+ behavioral and transactional features, and load scored results into **data warehouse** for downstream consumption.
- Built **model monitoring dashboards** in **Power BI** and **Tableau** to track real-time prediction accuracy, data drift, false positive rates, and business impact metrics; automated anomaly detection on model performance using **Python** and **CloudWatch**, reducing false-alert volume by **42%**.
- Established **continuous integration and deployment** workflows using **Jenkins** and **Git** to version-control model artifacts, retrain pipelines weekly, and promote models from development to production; achieved **99.7% uptime** and zero unplanned downtime over **18-month production cycle**.

SWIFT Payment Analytics & Anomaly Detection Dashboard (HCL Technologies)

- Designed a **Power BI analytics dashboard** integrating live **SWIFT MT/MX message flows** from enterprise payment systems, visualizing transaction volumes, message latencies, and ISO 20022 compliance metrics for regulatory and operational stakeholders; reduced manual reconciliation time by **55%**.
- Applied **statistical analysis** and **exploratory data analysis** techniques to identify anomalies in cross-border payment patterns, SEPA transaction flows, and international remittances; built **Python-based alerting logic** to flag outliers in **real-time**, catching **12 fraud cases** in first **6 months**.
- Automated **ETL workflows** using **Python** scripts and **SQL** queries to extract payment message data, normalize ISO 20022 fields, and load enriched datasets into **data warehouse**; optimized query performance by **38%** through indexing and query refactoring.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)